Paper IX

Eighth Lean Verification Block

Differential closure and higher-power standard-window bounds

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Scope and result

The all-residue Theorem 7.5 remains pending. This release proves its **full saturated higher-power estimate throughout the standard window**, rather than only a period modulo p. For every prime p, positive $m \mid p-1$, $0 \le j < p$, and natural n, k, u, put $h = (p-1)/m$. Then

$$\Delta_{uh}^{k} C_{mp,j}(n) = 0 \quad \text{or} \quad v_p\Big(\Delta_{uh}^{k} C_{mp,j}(n)\Big) \ge \min\{k, mn\}.$$

The unnormalized family $j!C_{mp,j}(n)$ satisfies the same bound. Both statements concern the original rational coefficient function, identified with its actual integral p-adic lift. They include $p = 2$, $k = 0$, $n = 0$, and the additional harmless case $u = 0$.

Coverage remains 23 of 24 named manuscript-proved items. Three classical inputs and two conjectures remain separate. The new result covers every permissible residue when $m = 1$, but not the range $p \le j < mp$ when $m > 1$.

New module	Theorems	Definitions
`DworkAlgebra`	7	3
`DworkClosure`	20	2
`HurwitzDworkClosure`	15	2
`ResidueZeroDwork`	9	2
`DifferenceStepMultiples`	5	1
`EulerDefect`	4	2
`StandardWindowDwork`	14	1
`StandardWindowPrecision`	5	0
`DworkWindowExamples`	8	0
Total added	87	13

The cumulative corpus has **526 explicit theorem declarations and 107 explicit definitions in 47 modules**. Counts include support lemmas and examples, not independent discoveries. All 38 upstream proof modules and seven earlier supplements are preserved.

1. The characteristic-zero product rule

The normalized defect is not an ordinary derivation

Let A be a commutative algebra over a characteristic-zero field K, and let U be an actual K-derivation. The source defines

$$B = \frac{U^{p^2} - U^p}{p}, \qquad M_q(f,g) = \sum_{a=1}^{q-1} \frac{1}{p}\binom{q}{a}(U^a f)(U^{q-a} g), \quad q = p, p^2.$$

The coefficients of each M_q are proved integral: the middle entries of a prime-power binomial row are divisible by p. In Lean they are implemented as natural quotients, with exact division proved before the product identity is used. Induction on derivative order establishes the full Leibniz formula and hence

$$\boxed{B(fg) = B(f)g + fB(g) + M_{p^2}(f,g) - M_p(f,g).}$$

The middle terms are essential. At $p = 3$, $f = t$ and $g = t^2$, the constant coefficient of $B(fg)$ is -2, whereas that of $B(f)g + fB(g)$ is zero. This inequality is itself Lean-checked.

All orders, without a circular closure assumption

Let $I \subset A$ be a subring stable under U. Define

$$\mathrm{Bounded}_r(f) \iff B^s f \in I \ (0 \le s \le r), \qquad \mathrm{Flat}(f) \iff B^s f \in I \ (s \ge 0).$$

The definitions contain no product or inverse closure requirement. Since B commutes with U, derivatives of a bounded element remain bounded. Induction on r in the displayed product rule then proves closure under multiplication. The finite middle sums use the lower-order induction hypothesis, not the desired next-order conclusion.

For an actual inverse $fg = 1$, applying the product rule and multiplying by g gives

$$B(g) = -g^2 B(f) - gM_{p^2}(f,g) + gM_p(f,g).$$

If $g \in I$, induction on r proves boundedness of g from boundedness of f. At each successor stage the preceding stage supplies boundedness of g needed on the right. Passing to all finite orders proves reciprocal closure for an integral inverse.

Principal targets.

`DworkAlgebra.B_mul`

`DworkClosure.bounded_mul, DworkClosure.bounded_inverse`

`DworkClosure.flat_mul, DworkClosure.flat_inverse`

2. The actual integral Hurwitz lattice

Connecting differential expressions to coefficient divisibility

The specialization works inside $\mathbb{Q}_p[[t]]$ and uses factorial-scaled coefficients

$$a_f(d) = d![t^d]f, \qquad I_p = \{f \in \mathbb{Q}_p[[t]] : a_f(d) \in \mathbb{Z}_p \text{ for every } d\}.$$

The source constructs I_p as a subring. Actual exponential-series multiplication gives

$$a_{fg}(d) = \sum_{r=0}^{d} \binom{d}{r} a_f(r)a_g(d-r), \qquad a_{Uf}(d) = a_f(d+1).$$

These identities prove multiplication and derivative stability of the lattice. Membership does not require ordinary power-series coefficients to lie in $\mathbb{Z}_p$.

The reciprocal is genuinely integral

For $f \in I_p$ with constant coefficient one, the usual power-series reciprocal is used. Its inverse equation yields the exact coefficient recurrence

$$a_{f^{-1}}(d) = \delta_{d0} - \sum_{r=0}^{d-1} \binom{d}{r} a_{f^{-1}}(r)a_f(d-r).$$

Strong induction proves that all of these coefficients are integral. Combining this construction with the preceding all-order theorem proves flatness of the *actual reciprocal*. No flatness or integrality of that reciprocal is assumed in the normalized-series specialization.

The earlier coefficient criterion is recovered exactly

Define the literal coefficient defect

$$(\delta a)(d) = a(d+p^2) - a(d+p).$$

The new source proves, for every s, d,

$$a_{B^s f}(d) = p^{-s}(\delta^s a_f)(d).$$

For an integral coefficient x, it proves the bridge

$$\boxed{p^{-s}x \in \mathbb{Z}_p \quad \Longleftrightarrow \quad p^s \mid x \text{ in } \mathbb{Z}_p.}$$

Thus the differential definition is equivalent to the existing coefficient-level Dwork criterion. Division occurs in $\mathbb{Q}_p$, while the assertion after division is membership in $\mathbb{Z}_p$. It is not divisibility in the field. Zero coefficients satisfy every finite precision condition.

Principal targets.

`HurwitzDworkClosure.integral_inverse`

`HurwitzDworkClosure.flat_product, HurwitzDworkClosure.flat_reciprocal`

`HurwitzDworkClosure.decode_B_iterate`

`HurwitzDworkClosure.normalized_mem_iff, HurwitzDworkClosure.flat_encode_iff`

3. A nonsingular Euler-operator argument

The standard interval without an analytic power-series-in-the-index premise

On Hurwitz coefficients, the actual Euler operator plus c is

$$(\Lambda_c a)(d) = (d + c)a(d).$$

Direct induction proves its commutator with every defect iterate:

$$\delta^s \Lambda_c a = \Lambda_c \delta^s a + sp\big(pU^{p^2} - U^p\big)\delta^{s-1} a, \qquad s \geq 1.$$

Suppose a is integral and supported at degrees divisible by p, and $p \nmid c$. Every supported pivot $d + c$ is proved to be a unit of $\mathbb{Z}_p$. The defect preserves this support. Induction in the commutator identity therefore proves

$$\mathrm{Flat}(\Lambda_c a) \Longrightarrow \mathrm{Flat}(a).$$

This is a reflection theorem on the integral supported domain. It does not assume flatness of the proposed preimage, and does not assert invertibility at a singular pivot $p \mid c$.

Apply it to the explicit normalized sections

Set $N = mp$ and, for $0 \leq j < p$, define integral Hurwitz coefficients

$$g_j(d) = \begin{cases} \binom{d+j}{d}^{-1}, & N \mid d, \\ 0, & N \nmid d. \end{cases}$$

The inverse binomial is constructed using the actual unit pivots already proved by Lucas reduction. Its image in $\mathbb{Q}_p$ is proved equal to the displayed reciprocal.

For $j = 0$, g_0 is the literal N-divisibility indicator. Because $N \mid p(p-1)$, the two shifts in δg_0 agree and $\delta g_0 = 0$. It is flat at every order. The actual binomial identity gives

$$\Lambda_{j+1} g_{j+1} = (j + 1)g_j.$$

As $1 \leq j + 1 < p$, the nonsingular reflection theorem inductively proves flatness of every g_j in the standard window. There is no uniform-convergence assumption in this argument.

Principal targets.

`EulerDefect.iterated_commutator, EulerDefect.flat_of_euler_flat`

`StandardWindowDwork.kernel_coe, StandardWindowDwork.kernel_euler_step`

`StandardWindowDwork.kernel_flat`

4. From these sections to the original Euler packets

Actual reciprocal, actual rational sequence, every order and step multiple

The factorial-ratio identity is proved coefficient by coefficient:

$$\sum_{d\geq 0} g_j(d)\frac{t^d}{d!} = j!\sum_{r\geq 0}\frac{t^{Nr}}{(Nr+j)!}.$$

The right side is precisely the normalized source kernel from the first block, not a replacement family. The earlier rational identity for its reciprocal is transported to $\mathbb{Q}_p[[t]]$. A separate encoding theorem identifies the reciprocal's coefficients with the already constructed $\mathbb{Z}_p$ lifts of the original $C_{N,j}(r)$.

All-order reciprocal closure therefore gives flatness of the actual packed Euler coefficient sequence. The earlier coefficient-extraction theorem now has a *proved antecedent* for every $j < p$, and yields

$$p^{\min(k,mn)} \mid \Delta_h^k C_{mp,j}(n) \quad \text{in } \mathbb{Z}_p, \qquad h = (p-1)/m.$$

A new finite telescoping argument proves the extension to arbitrary natural multiples of h. Its first-step identity is

$$f(n+uh) - f(n) = \sum_{a=0}^{u-1} \Delta_h f(n+ah).$$

Commutation of differences and induction on their order transfer any fixed ideal bound valid on a forward tail. The required precision $\min(k, mn)$ is nondecreasing as the starting row advances. Hence

$$\boxed{p^{\min(k,mn)} \mid \Delta_{uh}^k C_{mp,j}(n) \quad (0 \leq j < p).}$$

The scalar-extension bridge identifies these differences with the original rational coefficient function. The rational theorem explicitly allows zero differences, avoiding the library's finite valuation convention at zero. Since $j!$ is a unit for $j < p$, both normalized and unnormalized versions have the same precision.

Scope. The proof includes all primes, all rows and orders, and all natural step multipliers. For $m = 1$, the standard window is the complete allowed residue range. For larger m, residues $p \leq j < mp$ are not covered by this theorem. The earlier one-power period is not the source of the new higher-power estimate.

Principal targets.

`StandardWindowDwork.window_is_reciprocal, StandardWindowDwork.standard_window_flat`

`StandardWindowDwork.standard_window_global_bound`

`StandardWindowPrecision.normalized_window_precision`

`StandardWindowPrecision.unnormalized_window_precision`

`StandardWindowPrecision.modulus_p_complete_family`

5. Examples, verification and the remaining gap

The coverage count is unchanged; the unresolved theorem now has a stronger proved part

The original rational recurrence gives

$$C_{6,2}(1) = -\frac{1}{28}, \quad C_{6,2}(2) = \frac{425}{364}, \quad C_{6,2}(3) = -\frac{2039759}{2660}.$$

Consequently, the new source proves

$$\Delta_1^2 C_{6,2}(1) = -\frac{949959}{1235}, \qquad v_3\Big(\Delta_1^2 C_{6,2}(1)\Big) = 2.$$

This attains the bound $\min(2, 2 \cdot 1) = 2$. A claim of a third power is rejected. The separate cubic example proves that the normalized differential defect is not an ordinary derivation. Controls also reject an asserted unit pivot at $c = p$ and the prime-power binomial-divisibility formula at a composite base.

Recorded checks

The cumulative verifier rebuilds all 47 modules and audits all 633 explicit declarations: 526 theorems and 107 definitions. Warnings are errors, unexpected axiom dependencies are rejected, and 27 deliberately invalid proof controls are tested. The allowed transitive axioms remain `propext`, `Classical.choice` and `Quot.sound`. Exact results, compiler commands and source hashes are in the accompanying machine-readable reports.

All 189 infrastructure and release tests passed. Independent exact-arithmetic mirrors checked 3,040 normalized and 3,040 unnormalized bounds in 95 standard-window families, 6,596 source defects and 6,596 reciprocal defects, 75 product identities, 360 inverse-closure cases, 720 Euler commutators and 150 telescoping identities. They are finite Python mirrors, not code extracted from Lean and not substitutes for the universal proofs.

Failed development snapshots and diagnostics are preserved separately. Historical source files and earlier supplements remain unchanged. Mathlib's pinned compiled cache is reused, not independently rebuilt; the original historical numerical suite is not rerun. The manuscript-to-formal correspondence remains a mathematical review.

What is still unproved

The all-residue Theorem 7.5 requires the additional range $p \le j < mp$. Its supporting prose proof treats singular ratios at $c = p, 2p, \dots, (m-1)p$ through weighted Amice estimates and a dual Mahler pairing, then propagates across the intervening indices. Those actual-family estimates and the full remaining propagation are not supplied by this release. A unit argument at $p \nmid c$ cannot be used at $p \mid c$.

Release boundary: 23/24 named manuscript-proved items, with Theorem 7.5 still pending. The two conjectures remain unproved. The proved standard-window result is an infinite family, but no numerical proportion of the remaining work is inferred from the width of that window.